

Homework 4

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Exercise 1

Part (a)

We seek to solve the following system of linear congruences:

$$x \equiv 1 \pmod{2}$$

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 4 \pmod{7}$$

Let $M = 2 \cdot 3 \cdot 5 \cdot 7 = 210$. Then $M_1 = \frac{210}{2} = 105$, $M_2 = \frac{210}{3} = 70$, $M_3 = \frac{210}{5} = 42$, and $M_4 = \frac{210}{7} = 30$. We now solve the congruences:

$$105x_1 \equiv 1 \pmod{2}$$

$$70x_2 \equiv 1 \pmod{3}$$

$$42x_3 \equiv 1 \pmod{5}$$

$$30x_4 \equiv 1 \pmod{7}$$

Equivalently:

$$x_1 \equiv 1 \pmod{2}$$

$$x_2 \equiv 1 \pmod{3}$$

$$2x_3 \equiv 1 \pmod{5}$$

$$2x_4 \equiv 1 \pmod{7}$$

We see that we have the solution $(x_1, x_2, x_3, x_4) = (1, 1, 3, 4)$. Then:

$$x = 1 \cdot 105 \cdot 1 + 2 \cdot 70 \cdot 1 + 3 \cdot 42 \cdot 3 + 4 \cdot 30 \cdot 4 = 1103$$

Part (b)

We seek to solve the following system of linear congruences:

$$3x \equiv 2 \pmod{4}$$

$$4x \equiv 1 \pmod{5}$$

$$6x \equiv 3 \pmod{9}$$

Equivalently:

$$x \equiv 2 \pmod{4}$$

$$x \equiv 4 \pmod{5}$$

$$x \equiv 2 \pmod{3}$$

Let $M = 4 \cdot 5 \cdot 3 = 60$. Then $M_1 = \frac{60}{4} = 15$, $M_2 = \frac{60}{5} = 12$, and $M_3 = \frac{60}{3} = 20$. We now solve the congruences:

$$15x_1 \equiv 1 \pmod{4}$$

$$12x_2 \equiv 1 \pmod{5}$$

$$20x_3 \equiv 1 \pmod{3}$$

Equivalently:

$$3x_1 \equiv 1 \pmod{4}$$

$$2x_2 \equiv 1 \pmod{5}$$

$$2x_3 \equiv 1 \pmod{3}$$

We see that we have the solution $(x_1, x_2, x_3) = (3, 3, 2)$. Then:

$$x = 2 \cdot 15 \cdot 3 + 4 \cdot 12 \cdot 3 + 2 \cdot 20 \cdot 2 = 314$$

Part (c)

We seek to solve the following system of linear congruences:

$$x \equiv 3 \pmod{6}$$

$$x \equiv 7 \pmod{10}$$

$$x \equiv 9 \pmod{15}$$

However, note that $(10, 15) \nmid (9 - 7)$; therefore no solution exists.

Exercise 2

Part (a)

By Wilson's theorem, $30! \equiv (31 - 1)! \equiv 30 \pmod{31}$.

Part (b)

By Wilson's theorem, $22! \equiv (23 - 2)! \equiv 1 \pmod{23}$.

Part (c)

Note that $\frac{31!}{22!} \pmod{11}$ can be simplified as follows:

$$\begin{aligned}\frac{31!}{22!} &\equiv 31 \cdot 30 \cdot 29 \cdot 28 \cdot 27 \cdot 26 \cdot 25 \cdot 24 \cdot 23 \pmod{11} \\ &\equiv 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \pmod{11} \\ &\equiv 9! \pmod{11}\end{aligned}$$

By Wilson's theorem, $\frac{31!}{22!} \equiv 9! \equiv (11 - 2)! \equiv 1 \pmod{11}$.

Exercise 3

Part (a)

Let p be an odd prime number. By Wilson's theorem, $(p-1)! \equiv -1 \pmod{p}$. Furthermore:

$$\begin{aligned}
 (p-1)! &= \prod_{i=1}^{p-1} i \\
 &= \prod_{i=1}^{(p-1)/2} i \cdot \prod_{i=(p+1)/2}^{p-1} i \\
 &= \prod_{i=1}^{(p-1)/2} i \cdot \prod_{i=1}^{(p-1)/2} (p-i) \\
 &= \prod_{i=1}^{(p-1)/2} (i \cdot (p-i))
 \end{aligned}$$

Then:

$$\begin{aligned}
 (p-1)! &\equiv \prod_{i=1}^{(p-1)/2} (i \cdot (p-i)) \pmod{p} \\
 &\equiv \prod_{i=1}^{(p-1)/2} (i \cdot -i) \pmod{p} \\
 &\equiv (-1)^{(p-1)/2} \left(\prod_{i=1}^{(p-1)/2} i \right)^2 \pmod{p} \\
 -1 &\equiv (-1)^{(p-1)/2} \left(\left(\frac{p-1}{2} \right)! \right)^2 \pmod{p} \\
 -(-1)^{(p-1)/2} &\equiv \left(\left(\frac{p-1}{2} \right)! \right)^2 \pmod{p} \\
 (-1)^{(p+1)/2} &\equiv \left(\left(\frac{p-1}{2} \right)! \right)^2 \pmod{p}
 \end{aligned}$$

Part (b)

Suppose $p \equiv 1 \pmod{4}$. Then:

$$\begin{aligned} p &= 4k + 1 \\ p + 1 &= 4k + 2 \\ \frac{p+1}{2} &= 2k + 1 \end{aligned}$$

Then $\frac{p+1}{2} \equiv 1 \pmod{2}$, and so $(-1)^{(p+1)/2} = -1$. Then:

$$\equiv \left(\left(\frac{p-1}{2} \right)! \right)^2 \equiv -1 \pmod{p}$$

Consequently, $\left(\frac{p-1}{2} \right)!$ is a solution of the quadratic congruence $x^2 \equiv -1 \pmod{p}$.

Part (c)

Suppose $p \equiv 3 \pmod{4}$. Then:

$$\begin{aligned} p &= 4k + 3 \\ p + 1 &= 4k + 4 \\ \frac{p+1}{2} &= 2(k+2) \end{aligned}$$

Then $\frac{p+1}{2} \equiv 0 \pmod{2}$, and so $(-1)^{(p+1)/2} = 1$. Then:

$$\equiv \left(\left(\frac{p-1}{2} \right)! \right)^2 \equiv 1 \pmod{p}$$

Consequently, $\left(\frac{p-1}{2} \right)!$ is a solution of the quadratic congruence $x^2 \equiv 1 \pmod{p}$.